CE 4720 Microcomputer Architecture and Interfacing

Lab 2: A Wallace Tree Multiplier

Team Participation Requirement

Our lab work requires equal participation of all lab team members. When distributing the work on the lab assignments among the team members, make sure that everybody contributes to all stages of the assignment. Each member must contribute to the design and must have a portion of the Verilog code they can claim as the main authors, and all members must participate in running simulations, debugging as well as testing the final design on the DE2 boards. Similarly, all team members must actively participate in the preparation of the lab reports. Please read carefully Section 4 - Report - Appendix, which details how participation and authorship should be documented in the lab reports. Not following these reporting requirements will result in notable point reduction for the team.
Individuals reporting marginal contributions to the assignment will receive reduced points compared to the rest of the team. Please make sure to plan well and actively participate - that is the best way to learn and gain confidence and proficiency.

AI Use Policy

The use of AI/LLMs is permitted in lab assignments under the regulations defined in *AI Use Policy in CE4720* (available on Canvas in the Supplementary Resources module).

1. Objective

- Get familiar with the DE2 board.
- Understand basic combinational design constructs in Verilog.
- Practice instantiation of parameterized Verilog modules.
- Practice the use of Quartus II software for creating Verilog modules as well as their incorporation in the top-level schematic design file.

2. Tasks

Design an 8-bit x 8-bit unsigned integer multiplicator using a Wallace Tree and test its operation using the Quartus Simulation Wave Editor and the DE2 board.

1) Get familiar with DE2 and Quartus II 13.0sp1

a. Go through the manual for the DE2 board (available on Canvas - *Quartus and DE Board Guides*); familiarize yourself with FPGA pin assignments for the inputs and outputs available on the DE2 board (push buttons, switches, discrete LEDs, seven segment displays, etc.)

b. Use Quartus II software to:

- Create a simple Verilog module which has two single-bit inputs A and B, and an output C = AB (bit-wise AND of A and B). Use continuous assignment to implement the function.
- Create a symbol file for the Verilog module.
- Instantiate the Verilog module in a schematic file using the symbol file created in the previous step.
- Connect the module inputs and the output to respective I/O pins in the schematic design file and verify the module functionality using the waveform simulator. Test output values for all possible input combinations.
- Using the Quartus II Assignment Editor, tie A and B to two push buttons, and the output C to a discrete LED on the DE2 board. Compile the project and test it on the DE2 board. Make sure it works as desired. You can follow the example from DE2 Board & Quartus II Resources/QuartusII_Verilog_Design.pdf. Note that the Assignment Editor in our version of Quartus differs from the one discussed in this pdf. To set up signal-to-pin connections using our version of the Assignment Editor, refer to Quartus_II_Introduction-V13.pdf. To find the actual pin labels for your version of the DE2 Board (Cyclone II or Cyclone IV), refer to DE2_CycloneII_UserManual.pdf (starting from Page 27) or DE2_CycloneIV_UserManual.pdf (starting from Page 32), respectively.

2) Create an 8-bit x 8-bit Wallace Tree multiplier

The multiplier must be implemented as a Verilog module and its internal structure will consist of other instantiated Verilog modules. This module will not be the top-level entity module, it will be later incorporated as one of the modules in a top-level schematic design file.

a. The multiplier will operate as follows:

- It will take two unsigned 8-bit integers as an input.
- It will multiply them using a combination of a partial product generator, a Wallace tree comprising multiple Carry-Save Adders (CSA's), and a full parallel adder.

b. Design structure:

- Create a partial product generator Verilog module that will produce all partial products required for the Wallace Tree.
- Create a parameterized CSA Verilog module that will take three partial products as inputs and produce a Carry vector and a Save vector. The widths of the Carry and Save output buses will be determined from the bit widths of the input partial products. Assume that the three input partial products all have the same bit width and will be parameterized by a single parameter that can be overloaded during module instantiation.
- In the Wallace tree multiplier module, instantiate the partial product generator, multiple CSA modules with appropriate widths of the input and output buses, and the full adder, and wire them together to implement the multiplier.

Your design must optimize the Wallace tree structure to minimize the propagation delay of the cascaded CSA modules.

3) Create a 7-segment decoder Verilog module

The module will take in a 4-bit binary number and produce output signals that can be used to drive a 7-segment common-anode display and display the input value as a hexadecimal digit (e.g., for a 4-bit input "1111," the module will produce 7-segment signals displaying the digit "F."

4) Final system

- Use the schematic design in Quartus II as the top-level entity.
- Insert the Wallace Tree Verilog module and multiple 7-segment decoder modules in the schematic design.
- Interconnect the modules using wires/buses in the schematic design in a way that the two input values and the final product would all be displayed on the available 7-segment displays.
- Define system input and output pins and their mapping to the hardware pins on your DE2 boards. The multiplier inputs will be drawn from the 16 SW switches on the board.
- Make sure to refer in the Pin Assignment Editor to the correct hardware pin labels for your version of the DE2 board - Cyclone II and Cyclone IV use different labels for the same peripherals.

5) Run a Quartus II simulation

Run a Quartus II simulation of your design for the three test cases specified in the Check-Off section. Include the simulation results as commented screenshots in the Design section of your lab report.

6) Synthesize and test

Synthesize the system and test its operation on the DE2 board.

3. Check-Off

Demonstrate the following multiplications on the DE2 board. Convert the input values from decimal to binary (for input via switches); verify that the hexadecimal values displayed on the 7-segment displays correctly represent the input values and their resulting products.

Test cases:

1. 247 x 199
2. 234 x 211
3. 221 x 241

For check-off videos, besides demonstrating the multiplications, show the schematic-level design in Quartus and briefly comment on the function of each module and how they are interconnected.

4. Report

Follow the informal report format posted in our Canvas module. Each report should include the following:

Heading (the title of the lab, to whom the report is submitted, author list, the date it was submitted).

Summary (brief one or two paragraphs describing the objective and results).

Design:

- This section must detail all steps taken during the design of the system and its testing, including commented waveform simulation results from Quartus. The level of detail must be sufficient for an expert reader to be able to reproduce your design and evaluation procedure. Simply stating that the required functionality was implemented, without enough detail on how, would render the report useless for readers.
- Outline the overall structure of your system; include a screenshot of your schematic-level design from Quartus.
- Discuss the function and internal implementation of each of the modules in your design - including the modules used as building stones within the Wallace Tree module. Describe how you approached the implementation algorithmically, do not elaborate on your module source codes line-by-line.
- Present the simulation results for the checkoff test cases in the report in the form of waveforms from the simulator. Annotate the waveform screenshots electronically in your text processor software to make it easy for the reader to understand which signal track represents what and where each of the three test cases start in the simulation. Present all three test cases in a single waveform window.

Equipment List.

Conclusion and Discussion (extended version of the Summary with the team's subjective evaluation of the project, issues faced on the way, found workarounds, etc.).

References.

Appendix:

- Well commented Verilog source code; include a name of a student with each section of code they authored (in the comments).
- Summary of contributions - present a three-column table where the first column will list the team members, the second column their contributions (which sections of the Verilog code they wrote/co-wrote, involvement in other stages - simulation, debugging, testing on the board, report writing; note that ALL team members must equally participate in the design stage and report writing) and the third column will present the total time spent by each member on the project.

Note: Task 1.b is intended only for initial practice with QII/DE2 and **should not** be discussed in the lab report.

5. Submission

Lab report in PDF format; submit one report per team (include names of all team members)

6. Dates

- Check-off due date is Thursday, April 2.
- Lab report due: Friday, April 10, 11:59 pm.

7. Student Checklist for Report & Deliverables

Use this checklist to make sure your Lab 2 submission meets all requirements. Each item may be graded, and missing pieces can cost points.

Category	Checklist Items
Wallace Tree Implementation	☐ Partial product generator Verilog module created and produces all required partial products. ☐ Parameterized CSA module created with correct inputs (3 partial products) and outputs (Carry and Save vectors). ☐ CSA module uses parameter for bit width that can be overloaded during instantiation. ☐ Wallace tree structure is optimized to minimize propagation delay of cascaded CSA modules. ☐ Full parallel adder included to produce final 16-bit product. ☐ All CSA modules are properly instantiated with appropriate bus widths.
7-Segment Decoder	☐ 7-segment decoder module takes 4-bit binary input. ☐ Decoder outputs correctly drive common-anode 7-segment display. ☐ All hexadecimal digits (0-F) display correctly.
Top-Level Integration	☐ Schematic design used as top-level entity. ☐ Wallace Tree module properly inserted in schematic. ☐ Multiple 7-segment decoder modules instantiated for displaying inputs and product. ☐ Two 8-bit input values displayed on 7-segment displays. ☐ 16-bit product displayed on 7-segment displays. ☐ 16 SW switches mapped correctly to multiplier inputs. ☐ Correct pin assignments used for DE2 board version (Cyclone II or Cyclone IV).
Simulation & Testing	☐ Quartus II simulation run for all three check-off test cases (247x199, 234x211, 221x241). ☐ Waveform screenshots included in Design section with annotations. ☐ All three test cases presented in a single waveform window. ☐ Signals clearly labeled and test case boundaries indicated.

Category	Checklist Items
	☐ Simulation results show correct products for all test cases. ☐ Design successfully synthesized and tested on DE2 board.
Check-Off Readiness	☐ Can demonstrate all three test cases on the DE2 board. ☐ Can convert decimal inputs to binary for switch input. ☐ Can verify hexadecimal displays match input values and products. ☐ Can show schematic-level design in Quartus. ☐ Can explain function of each module and their interconnections. ☐ Can explain how the Wallace tree reduces partial products using CSAs.
Report Structure & Quality	☐ Report follows the informal report template/structure required for the course (Heading, Summary, Design, Equipment List, Conclusion/Discussion, References, Appendix). ☐ Summary states the objective and key results (Wallace Tree multiplier implementation). ☐ Design section describes approach clearly enough that a knowledgeable reader could reproduce the design without seeing the code. ☐ Screenshot of schematic-level design included from Quartus. ☐ Function and internal implementation of each module discussed (algorithmic approach, not line-by-line code). ☐ Screenshots are clear, readable, and referenced in the text. ☐ Waveform screenshots include all three test cases with annotations. ☐ References section lists any external sources used (including Quartus/DE2 resources, textbooks/notes, and any AI tools if used).
Appendix & Team Contributions	☐ Appendix includes the complete, well-commented Verilog source code. ☐ Code comments indicate student authorship for major sections (who wrote/co-wrote which parts). ☐ Appendix includes a contributions table: Team member | Contributions | Total time spent. ☐ Contributions indicate that all members participated in design, simulation/debugging, and report writing (as required).
AI Use Disclosure (if	☐ AI/LLM use follows the CE4720 AI Use Policy posted on Canvas.

Category	Checklist Items
applicable)	☐ If an AI tool was used (syntax help, debugging ideas, proofreading, etc.), the report clearly discloses it (tool name + what it was used for). ☐ AI was not used to produce large copy-pasted blocks of final Verilog code or report text.
Final Submission	☐ One PDF report submitted to Canvas by the due date (one per team, with all team member names). ☐ Check-off completed by the due date (in-person or approved video process for remote/PEP teams). ☐ File names are sensible and match course expectations (easy to identify the team and lab).